

MQTT - configuration settings

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Date	Description
12.11.2024 - 0.1.0	• Creation of the document
29.11.2024 - 0.1.1	• Added FAQ • Fix. table names • Lower temperature ranges • Added temperature warning range • Changed to OR and && to AND • Added page numbers • Change F (force) to A (current) • Added picture for zones notation • Added refresh rate for sensors topics - 2Hz • Added lubrication parameters • Added force linearization equation
3.12.2024 0.1.2	• Reduce time from 300s to 30s - lubrication delay • added topics response from Digital Twin for #4 actuators ◦ <i>/DT/response/actuator#</i> • Change current for overcurrent to $I > 7,10A$: overcurrent •

Note: This documentation should be reviewed and updated regularly to reflect any changes in the MQTT broker configuration, sensor data configuration or topic structure.

Settings

MQTT

MQTT data:

MQTT broker address	tcp://192.168.1.1
Security MQTT username MQTT password	 / /
Port	1883 (default MQTT port)

All available topics

topic for machine

/haulick-32t

topics for tool

/haulick-32t/32777

topics for actuators

/haulick-32t/32777/actuator_1

/haulick-32t/32777/actuator_2

/haulick-32t/32777/actuator_3

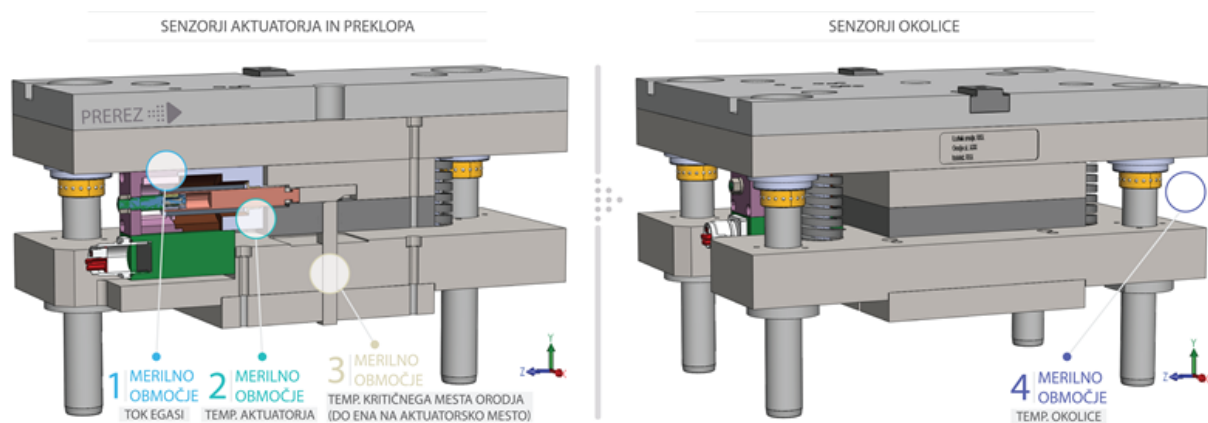
/haulick-32t/32777/actuator_4

Sensors

Sensors data for digital twin:

Name : Type	#	Zone	Description
Tool sensors	4	3	Measuring tool temperature (near important mechanical parts)
Environment sensor	1	4	Environment temperature
Actuator#1 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#2 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#3 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator
Actuator#4 sensors			
case sensor	1	2	Temperature - case of the actuator
current sensor	1	1	Current for actuator

*Topics are send at constant interval: 2Hz [can be adjusted]



Lubrication parameters for Haulick 32T with Raziol lubrication unit

1. **Lubrication Trigger:**
 - Triggered by a **rising edge** signal.
2. **Lubrication Time (t_lubrication):**
 - Fixed lubrication duration at the machine: **500 ms**.
3. **Lubrication Delay (t_lubrication_delay):**
 - Minimum time between consecutive lubrication requests: **30 seconds**.
4. **Lubrication Try-Out Limit (lubrication_tryOut):**
 - Maximum number of lubrication requests before triggering an error: **3 attempts**.



Sensor limits

50N configuration - temperature range

Sensor	Measuring range	Result
Case sensor	$T > 45^{\circ}\text{C}$ $T > 40^{\circ}\text{C}$ AND $T \leq 45^{\circ}\text{C}$ $T > 150^{\circ}\text{C}$ OR $T \leq 0^{\circ}\text{C}$ $T > 0^{\circ}\text{C}$ AND $T < 150^{\circ}\text{C}$	Error: overheating Warning: temperature Not connected Connected range
Current sensor / Force sensor*	$I > 7,10\text{A}$ $I > 5,39\text{A}$ AND $I \leq 7,10\text{A}$ $I > 1,92\text{A}$ AND $I \leq 5,39\text{A}$ $I > 4,53\text{A}$ AND $I \leq 5,39\text{A}$	Error: overrange Warning: current Working range Lubrication required
Tool sensor & Environment sensor	$\Delta t (\text{Tool} - \text{Environment}) \geq 5^{\circ}$ $5^{\circ} < \Delta t (\text{Tool} - \text{Environment}) > 2^{\circ}$ $-2^{\circ} < \Delta t (\text{Tool} - \text{Environment}) \leq 2^{\circ}$ $\Delta t < -2^{\circ}\text{C}$ $\Delta t > 2^{\circ}\text{C}$	Error: overheating Warning temp. Working condition Warning environment Warning environment

*Force-Linearization : $F = 11,71 \cdot I - 3,0$ [N]

50N configuration - times to final temperature

Speed	Environment temperature	Egasi temperature	Actuator case	Reaction time to final temperature
100RPM->150RPM	22.4 -> 23,5	30 -> 35	22.4->23,8	50 min
150RPM-> 200RPM	24.2 ->25,2	35 -> 35	24.2 ->25,8	65 min
200RPM->250RPM	25.2->25.5	35->40	25.2 ->26,9	7 min
250RPM -> 0RPM	27.2->27.6	40->30	29.3->28.2	20 min
0RPM -> 250RPM	27.7-> 27.5	30->40	28.2->29.3	19 min

TOPICS SETTINGS

Topic structure

topic: **/machine**

Main machine status topic

- Topic: */haulick-32t*
- Description: Root topic for Haulick 32T machine status
- Data Format: JSON

example for the topic:

/haulick-32t		
T_env	string	Environment temperature [°C]
is_in_continuous_mode	bool	Machine in continuous mode
is_in_error	bool	Machine has an error
speed	int	Machine speed [RPM]
current_count	int	Machine counter

Example payload:

```
{ "T_env": 23.5, "is_in_continuous_mode": true, "is_in_error": false, "speed": 150, "current_count": 251000 }
```

topic : /machine/**tool-nr**

Toll-nr machine status topic

- Topic: */haulick-32t/32777/*
- Description: Root topic for tool status
- Data Format: JSON

example for the topic:

<i>/haulick-32t/32777/</i>		
T_tool	int	Tool temperature [°C]

Example payload:

```
{ "T_tool": 23.5 }
```

topic /machine/tool-nr/**type-of-sensor_#number**

Actuator/motor status topic

- Topics:
 - /haulick-32t/32777/actuator_1,
 - /haulick-32t/32777/actuator_2,
 - /haulick-32t/32777/actuator_3,
 - /haulick-32t/32777/actuator_4
- Description: Root topic for actuator/motor status
- Data Format: JSON

example for the topic:

/haulick-32t/32712/actuator_1		
T	int	actuator temperature [°C]
Imax	int	actuator max. current [A]
Imin	int	actuator min. current [A]
Fmax	int	actuator max. force [N]
Fmin	int	actuator min. force [N]

Example payload:

{ "T": 23.5, "Imax":2.5, "Imin":-1A, "Fmax":50N, "Fmin":-10N }

Digital twin:

Same topic is for four other actuators (motors) example:

- /haulick-32t/32777/actuator_1 - motor #1
- /haulick-32t/32777/actuator_2 - motor #2
- /haulick-32t/32777/actuator_3 - motor #3
- /haulick-32t/32777/actuator_4 - motor #4

Response from DT (Digital Twin)

Response topics

Response status topic

- Topics:
 - */DT/response/actuator# - example /DT/response/actuator1, /DT/response/actuator2, /DT/response/actuator3, /DT/response/actuator4*
- Description: Root topic from digital twin status
- Data Format: JSON

example for the topic:

<i>/DT/response/actuator#</i>	Data type	Comment
req_lubrication	bool	request lubrication for actuator
is_current_error	bool	error flag
is_current_warning	string	warning flag
is_temeprature_error	bool	error flag
is_temperature_warning	bool	warning flag
T_sensor_disconnected	bool	flag for sensor disconnected
I_sensor_disconnected	bool	flag for sensor disconnected

example for the topic

<i>/DT/response/sensors/temperature</i>		
is_overheated	bool	request lubrication
is_in_range	bool	error flag from digital twin
T_sensor_disconnected	bool	error name - comment

FAQ